Week 4:
Decision Tree

WIA1006 | Semester 2 | Machine Learning | Tutorial Session | University of Malaya

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Questions and Solutions

Question #1

1. **What is ID3 and how it builds the decision tree?**

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Ans: ID3 is **a mathematical algorithm for building the decision tree**. ID3 uses **Entropy** and **Information Gain** as attribute selection measures **to construct a Decision Tree**.

A Decision Tree is built top-down from a root node and involves the partitioning of data into homogeneous subsets. **To check the homogeneity of a sample, ID3 uses entropy**. Therefore, entropy is zero when the sample is completely homogeneous, and entropy of one when the sample is equally divided between different classes.

Information Gain is based on the **decrease in entropy** after splitting a dataset based on an attribute. The meaning of **constructing a Decision Tree is** all about **finding the attributes having the highest information gain**.

Start with all training data

Compute entropy (how mixed the labels are)

Compute information gain for each attribute

Choose the attribute with highest information gain

Split the data and repeat on each branch

Key idea

Entropy measures impurity. Information gain measures how much a split reduces that impurity.

Tip:

- Mixed classes -> high entropy
- Pure groups -> low entropy
- Best split = biggest reduction in uncertainty

2. List down the different types of nodes in Decision Trees.

1. **Root node**: It is the top-most node of the Tree from where the Tree starts.
2. **Decision nodes**: One or more decision nodes that result in the splitting of data into multiple data segments. Our main goal is to have the children nodes with maximum homogeneity or purity.
3. **Leaf nodes**: These nodes represent the data section having the highest homogeneity.

Root node

Top-most node
First question asked

Decision node

Internal node
Splits data again

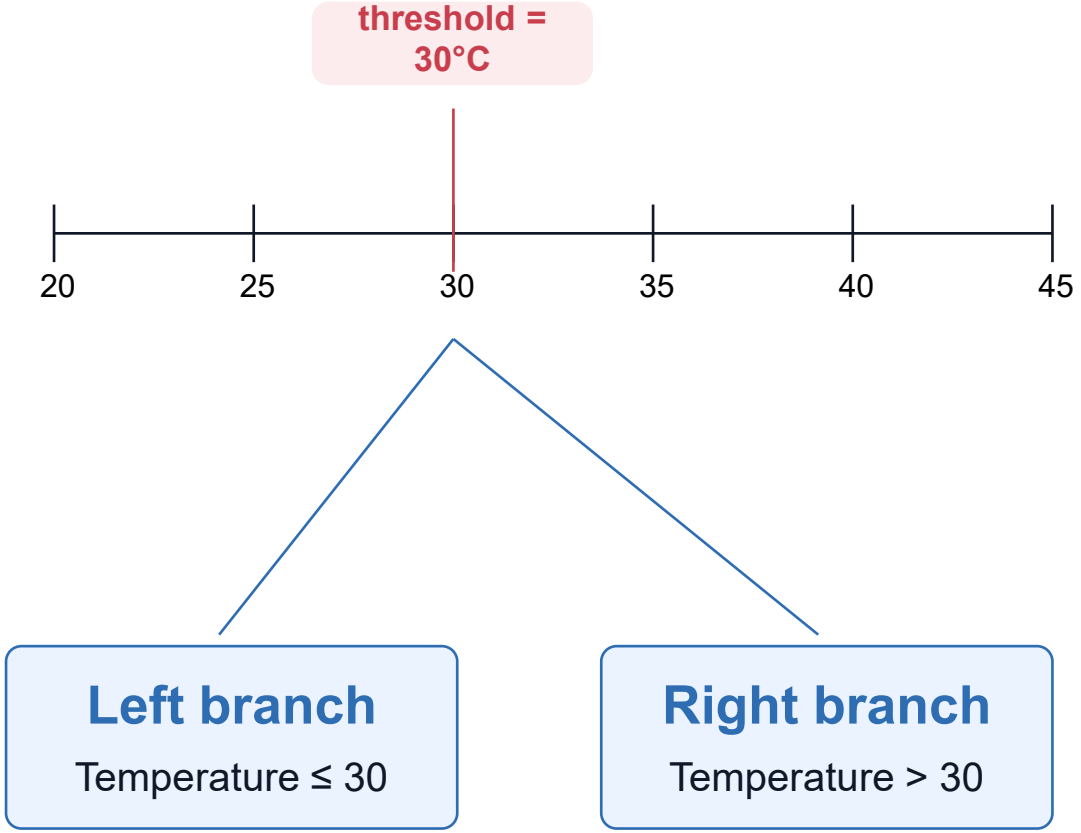
Leaf node

Final answer
Most homogeneous group

3. How Decision Tree handles continuous features?

Ans: Continuous (real-valued) features can be handled by allowing nodes to split a real valued feature into two ranges **based on a threshold** (e.g. length < 5 and length ≥ 35)

Continuous feature split



4. What do you understand by Pruning in a Decision Tree? Explain two common methods of Pruning.

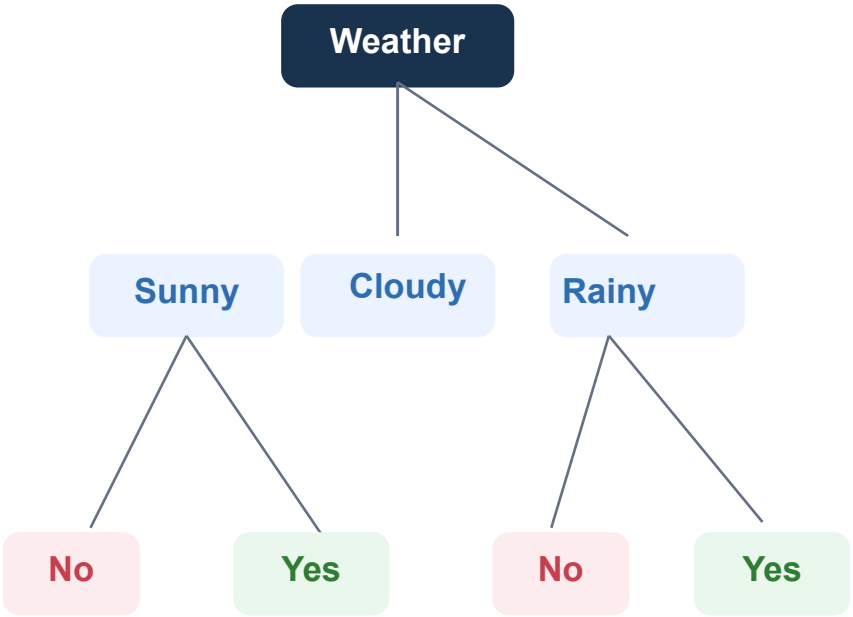
Ans: The process **of removing sub-nodes of a Decision node** is called pruning, which is the opposite process of splitting.

The two most widely used techniques for pruning are **Post** and **Pre-Pruning**.

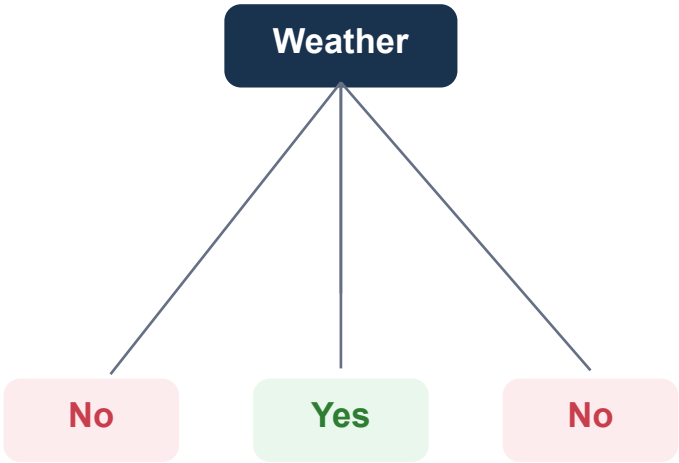
Prepruning: Stop growing tree as some point during top-down construction when there is no longer sufficient data to make reliable decisions.

Postpruning: Grow the full tree, then remove subtrees that do not have sufficient evidence.

Pre-Pruning



Post-Pruning



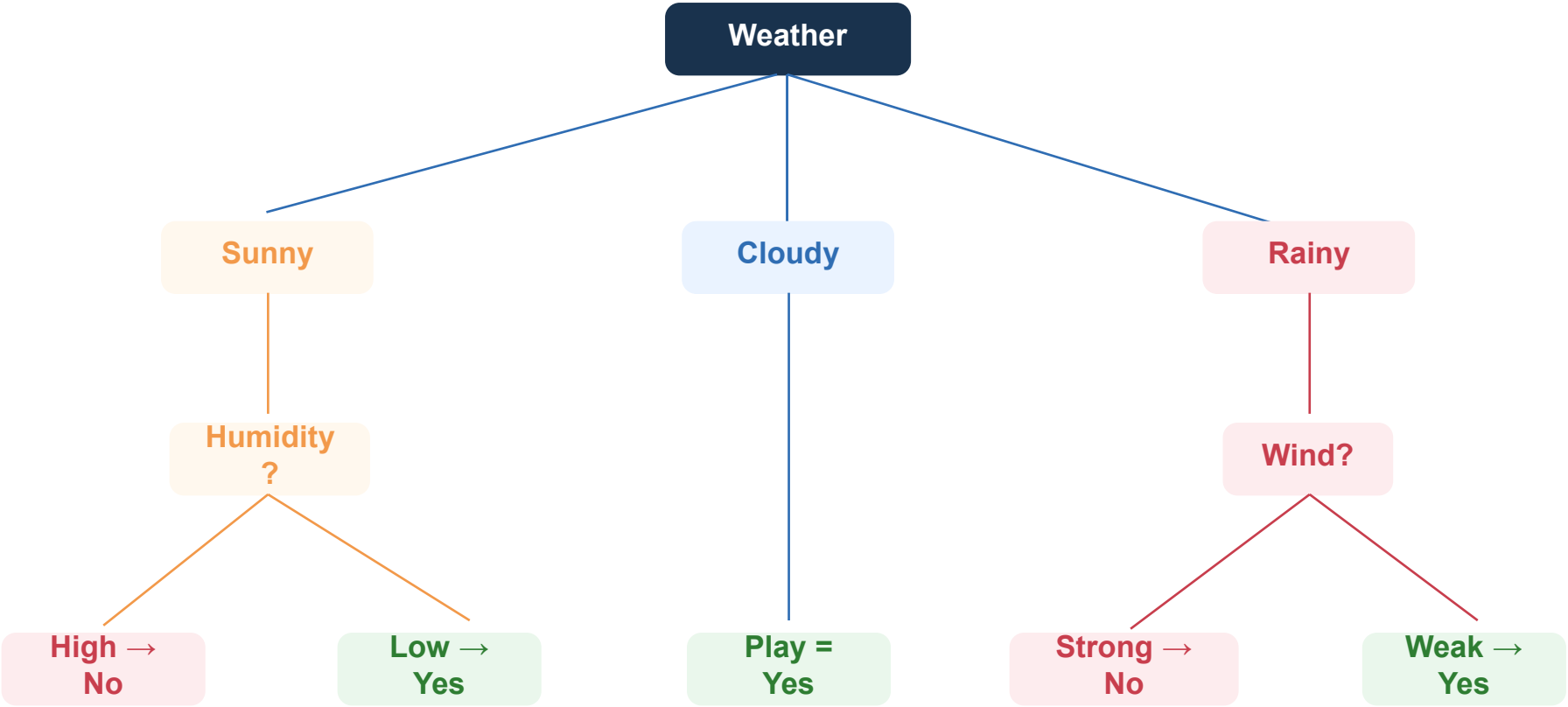
Pre-Pruning
Stop the growth early when the evidence is too weak to justify another split.

Post-Pruning
Grow the tree first, then remove branches that do not improve generalization.

5. Let's assume we want to play badminton on a particular day (say Saturday) how will you decide whether to play or not. Let's say you go out and check if it's hot or cold, check the speed of the wind and humidity, how the weather is, i.e. is it sunny, cloudy, or rainy. You take all these factors into account to decide if you want to play or not. The Observations of the last ten days are depicted in the below table.

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

i. Draw the Decision Tree for the concept play Badminton (Hint: Weather is the root node).



Tip: A decision tree mimics everyday reasoning - start with the most informative question, then narrow down to a final Yes/No decision.

ii. In part i, the Decision Tree assigns a Boolean classification (Yes or No, only 2 discrete values) to each example. Do you think Decision Trees are good practices when we are having discrete output values?

Ans: Yes, decision tree methods easily extend to learning functions with more than two possible output values. A more substantial extension allows learning target functions with real valued outputs, though the application of decision trees in this setting is less common.

6. We will use the dataset below to learn a decision tree which predicts if people pass machine learning (Yes or No), based on their previous GPA (High, Medium, or Low) and whether or not they studied. (For this problem, you can write your answers using log_2, but it may be helpful to note that $log_2 3 \approx 1.6$.)

GPA	Studied	Passed
L	F	F
L	T	T
M	F	F
M	T	T
H	F	T
H	T	T

Start with all training data

Compute entropy (how mixed the labels are)

Compute information gain for each attribute

Choose the attribute with highest information gain

Split the data and repeat on each branch

GPA	Studied	Passed
L	F	F
L	T	T
M	F	F
M	T	T
H	F	T
H	T	T

Start with all training data

Compute entropy (how mixed the labels are)

Compute information gain for each attribute

Choose the attribute with highest information gain

Split the data and repeat on each branch

a) What is the entropy H (Passed)?

Ans: H(Passed) = −(2/6 log_2 2/6 + 4/6 log_2 4/6)
H(Passed) = −(1/3 $log_2$1/3 + 2/3 $log_2$2/3)
H(Passed) = $log_2$3 – 2/3 ≈ 0.92

GPA	Studied	Passed
L	F	F
L	T	T
M	F	F
M	T	T
H	F	T
H	T	T

Dataset: 6 students

Passed = 4 | Failed = 2

$$H(Passed) = -\left(\frac{2}{6}\log_2\frac{2}{6} + \frac{4}{6}\log_2\frac{4}{6}\right) \approx 0.92$$

P P P P

F F

Mixed classes -> entropy is not zero

Entropy = 0
all labels same

Entropy = high
labels mixed

Why it matters
ID3 wants splits that lower entropy

Start with all training data

Compute entropy (how mixed the labels are)

Compute information gain for each attribute

Choose the attribute with highest information gain

Split the data and repeat on each branch

b) What is the Entropy and Information Gain for H (Passed | GPA)?

Ans: H(Passed|GPA) = − 1/3 (1/2 $log_2$1/2 + 1/2 $log_2$1/2) – 1/3 (1/2 $log_2$1/2 + 1/2 $log_2$1/2) – 1/3 (1 $log_2$1)

H(Passed|GPA) = 1/3 (1) + 1/3 (1) + 1/3 (0)

H(Passed|GPA) = 2/3 ≈ 0.66

IG(Passed|GPA)= 0.92 – 0.66 = 0.26

Start with all training data

Compute entropy (how mixed the labels are)

Compute information gain for each attribute

Choose the attribute with highest information gain

Split the data and repeat on each branch

c) What is the Entropy and Information Gain H (Passed | Studied)?

Ans: H(Passed|Studied) = − 1/2 (1/3 $log_2$1/3 + 2/3 $log_2$2/3) – 1/2 (1 $log_2$1)

H(Passed|Studied) = 1/2 log2 3 – 1/3 ≈ 0.46

IG(Passed|Studied)= 0.92 – 0.46 = 0.46

Start with all training data

Compute entropy (how mixed the labels are)

Compute information gain for each attribute

Choose the attribute with highest information gain

Split the data and repeat on each branch

Original Target

H(Passed) ≈ 0.92

Conditional Entropy

How mixed the groups remain after a split

Information Gain

How much uncertainty is removed by the split

Split by GPA

H(Passed | GPA) ≈ 0.66
IG(Passed | GPA) = 0.92 - 0.66 = 0.26

Split by Studied

H(Passed | Studied) ≈ 0.46
IG(Passed | Studied) = 0.92 - 0.46 = 0.46

Using the values in the above: Studied has higher Information Gain (0.46) than GPA (0.26), so Split by Studied would be chosen first.

Record your attendance

Thank You!

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